MILITARY SPECIFICATION

LIFEBOAT, MAN OVERBOARD, CO_2 INFLATABLE

1. SCOPE

1.1 Scope. - This specification covers a lifeboat, man overboard, CO_2 inflatable type, complete with stowage case for use in man overboard recove ery.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
CCC-T-191 - Textile Test Methods
PPP-B-636 - Boxes, Fiberboard

MILITARY
MIL-A-7021 - Asbestos Sheet, Compressed, for Fuel, Lubricant, Coolant, Water, and High Temperature Resistant Gaskets
MIL-L-7178 - Lacquer, Cellulose Nitrate, Gloss, for Aircraft Use
MIL-C-7905 - Cylinders, Compressed Gas, Nonshatterable
MIL-P-8585 - Primer Coating, Zinc Chromate, Low-Moisture Sensitivity
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-12647 - Pumps, Inflating, Deflating, Hand-Operated, Air, Single-Cylinder, Plunger Type
MIL-R-13689 - Reflectorized Sheeting, Adhesive (Retro-Reflective)
MIL-C-17415 - Cloth, Coated, and Webbing, Inflatable Boat and Miscellaneous Use
MIL-I-18369 - Inflation Equipment, Carbon Dioxide for Multiplace Liferafts
MIL-L-22686 - Light for Inflatable Boats

STANDARDS

FEDERAL
FED-STD-595 - Colors

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

AIR FORCE - NAVY AERONAUTICAL
AN-227 - Fasteners, Snap and Curtain
AN-6025 - Cylinder, Nonshatterable Compressed Gas

BUREAU OF SHIPS
645-1428951 - Hoisting, Towing and Miscellaneous Fittings for Inflatable Boats
645-1973948 - Stowage Case for Man Overboard Boat
S8225-1298540 - Sea Anchor for Inflatable Boats
S8225(7)-1428724 - Man Overboard Boat CO_2 Inflatable, General Arrangement

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION
Handbook H28 - Screw Thread Standards for Federal Services

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

FSC 1940

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3. REQUIREMENTS

3.1 Qualification. - The lifeboats furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein and has been listed on or approved for listing on the applicable qualified products list.

3.2 Preproduction sample. - Before production is started, a preproduction sample shall be submitted for tests specified in 4.7 to determine conformance with this specification.

3.3 Material. - Material shall be as specified herein and as shown on the applicable drawings. Material not definitely specified shall be of the best commercial quality, of the lightest possible weight, and shall be entirely suitable for the purpose intended. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.3.1 Government furnished material. - Two life preservers will be furnished by the Government for each lifeboat.

3.3.2 Coated cloth. - The coated cloth used in the construction of the boat shall conform to Specification MIL-C-17415 except that the color shall be orange between textile sea rescue color and Standard FED-STD-595 color No. 12197 and shall be of the following types and classes:

Part	Type	Class
Bottom	16	A
Bulkheads	16	B
Arch tube	16	A
Light attachment patches and light holder	16	A
CO_2 cylinder holders	16	A
"Dee" ring patches	16	A
Inflatable floor	1, 6 (see 3.6.6)	
Life line patches	16	A
Main tube	16	A
Pockets	16	A
Protection liners	16	B
Tape	8	A or B
Water ballast pockets	16	A
Righting and boarding handles	16	A

3.3.2.1 Utilization of scrap coated cloth. - The use of scrap coated cloth for the fabrication of patches, pockets and flaps, cylinder holders and life line supports is acceptable, and it shall be employed whenever practicable. However, wrinkled or inferior quality scrap coated cloth will not be permitted for any purpose.

3.3.3 Color. - The boat shall be orange color between textile sea rescue color and Standard FED-STD-595 color No. 12197.

3.3.4 "Dee" rings. - The "Dee" rings and sleeves shall be in accordance with Drawing 645-1428951.

3.3.5 Fasteners. - Snap fasteners shall be Marlex or equal (high density polyethylene) and shall be in accordance with Drawing AN-227.

3.3.6 Finish. - Aluminum alloy parts shall be anodically treated in accordance with Specification MIL-A-8625. Corrosion-resistant steel parts shall be either sand blasted or left unfinished if these parts do not come in contact with other dissimilar metals. All exposed metal surfaces of the cylinder valves shall be painted with one coat of zinc chromate primer conforming to Specification MIL-P-8585, and two coats of light blue lacquer conforming to Specification MIL-L-7178. All other metal parts, not specifically mentioned, shall be of a noncorrosive material or suitably coated to prevent rust or corrosion.

3.3.7 Rope and line. - All rope and line shall be nylon. One-eighth inch diameter line shall have a minimum breaking strength of 300 pounds and 1/4 inch diameter rope shall have a minimum breaking strength of 1200 pounds. Braided line shall be diamond or solid braid.

Part	Diameter (inch)
Life line	1/4
Ballast pocket trip line (braided)	1/8
Stowage case plug line (braided)	1/8
Life preserver attachment line (braided)	1/8

3.3.7.1 The ends of all nylon lines shall be smoothly fused.

3.3.8 Reflective tape. - The reflective tape shall be white, 2 inches wide in accordance with Specification MIL-R-13689 and installed as shown on Drawing S8225(7)-1428724 attached with Minnesota Mining and Manufacturing Company FE-52 bonding cement or equal.

3.3.9 Inflation system. -

3.3.9.1 Cylinders. - Cylinders shall be 96 cubic inch capacity and shall conform to type B of Drawing AN-6025 in accordance with Specification MIL-C-7905,

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except that wire wrapping shall not be used and the gunfire test shall require only that the cylinder shall remain in one piece. Tumbling of the shot is not required.

3.3.9.2 The flexible hose connection between the CO_2 cylinder valves and the manifolds shall be in accordance with Drawing 645-1428951.

3.3.9.3 Hand pumps. - One plastic 6-inch stroke hand pump with a 8 inch long hose shall be provided for topping-off and deflating. The hand pump shall conform to type II, size 1, class A of Specification MIL-P-12647 having a male fitting, 13/16 inch diameter, 20 threads per inch. The pump shall be furnished with an adapter to fit the inflation valve and hose. The word "Inflator" shall be conspicuously marked on the discharge end of the pump and the word "Deflator" conspicuously marked on the suction end. One additional adapter shall be stowed with the repair items. The pump, hose with adapter attached, as packaged, shall be heat sealed in a 4 mil thick polyethylene container. Included air volume shall be kept to a practical minimum. Sealed package to be placed in the pump pocket.

3.3.10 Thread. - All thread shall be nylon except as required by 3.7.3.

3.3.11 Radar reflective screen. - The radar reflective screen shall be made of metallized (deposited silver) nylon mesh coated, orange color between textile sea rescue color and Standard FED-STD-595 color No. 12197, in such a manner as to be waterproof with the interstices open. The screen shall be bound with 1 inch grommetted nylon tape. The grommetts shall be brass, size No. 0. One-eighth inch diameter nylon line shall be used for securing the bottom of the screen to the loops fitted on the inflatable floor and main tube. The longitudinal screen shall be attached to the arch tube by brass and molded high density polyethylene, Carr Fastener Corporation parts number X2/10128, PC10380, PC10223 and BS10412 or equal. The athwartship screen shall be sewn at right angles to the longitudinal screen. The longitudinal screen is to be secured in place while the boat is inflated, the athwartship screen is to be left loose.

3.4 Component parts. - The completed boat shall include the boat itself with the equipment specified herein.

3.5 Weight. - The weight of the completed boat shall be kept to the minimum.

3.6 Manufacture and assembly. -

3.6.1 Dimensions and shape. - The dimensions and shape of the lifeboat shall conform to Drawing S8225(7)-1428724.

3.6.2 Laps and seams. - All laps and seams shall be secured only by use of vulcanizing cement. Sewing shall not be used in any seams or through the coated cloth of any air chamber. Sewing will be permitted only in the construction of patches, disks, flaps, life line supports, CO_2 cylinder holders and pockets, but not in their final attachment to the lifeboat. Vulcanizing in a pressure type autoclave shall be required. The minimum overlap of seams shall be 3/4 inch. The amount of lap shall be sufficient to make the seams as strong as the coated cloth joined and the seams shall be required to withstand a test load equal to the coated cloth tensile strength. Vulcanizing cement shall be applied in a line parallel to the edge of the lap, tape or gum stripping and shall extend approximately 1/8 inch beyond the tape or gum stripping edges. No two laps or seams shall coincide. Seams shall be covered inside and outside with tape at least 1-5/8 inches wide after application and conforming to type 8, class B of Specification MIL-C-17415, or gum stripping, having a gauge thickness of 0.032 ± 0.005 inch and having a minimum width of 3/4 inch after application. Tape or gum stripping may be omitted on the outside of a lap where another layer of fabric covers the lap. Tape or gum stripping shall be applied without excessive tension, and shall be so laid that an equal amount is on each side of the seam edge which it covers.

3.6.3 Bulkhead. - The main tube shall be fitted with internal vertical bulkheads, dividing the flotation tube into two separate compartments. These bulkheads shall be of either hemispherical or conical design. Bulkheads shall be located approximately as shown on Drawing S8225(7)-1428724 in order that the volume of the two compartments will be approximately equal. The construction of the bulkheads and the security attachments shall be such that when one compartment only is inflated to 2 pounds per square inch (p.s.i.) (the other compartment being deflated), there shall be no failure when tested in accordance with 4.7.4. The bulkhead coated cloth shall be flanged and attached directly to the tube or protection liner and shall also be shoe taped with coated cloth tape at least 1-5/8 inches wide, in the opposite direction, as shown on Drawing S8225(7)-1428724. The bulkhead may be covered, each side, with a layer of natural rubber gum at the option of the contractor.

3.6.3.1 Protection liners. - Protection liners shall be fitted as shown on Drawing S8225(7)-1428724 on the inside of the tubes to prevent damage to the tube fabric when CO_2 is released. Liners are to be attached only at bulkheads, manifolds, ends and as necessary to aid in assembly.

3.6.4 Main tube. - The main tube shall be constructed as shown on Drawing S8225(7)-1428724. The main tube shall be capable of withstanding a minimum

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internal pressure of 5 p.s.i. Boarding handles shall be fitted.

3.6.5 Bottom. - A bottom applied without tension, shall be attached to the main tube as shown on Drawing S8225(7)-1428724. The attachment shall be reinforced by a shoe of two plies of type 8, class B tape fabric in accordance with Specification MIL-C-17415, 2 inches wide folded in a V-shape with 1 inch attached to the bottom and 1 inch attached to the tube, along the inside tangent line around the entire peripheral distance. Tape coated cloth conforming to type 16, class A requirements of Specification MIL-C-17415, 1-5/8 inches wide, shall be cemented over the outer edge of the bottom so as to be equally divided between the bottom coated cloth and the tube coated cloth. A righting handle shall be fitted as shown on Drawing S8225(7)-1428724.

3.6.6 Inflatable floor. - An inflatable floor, as shown on Drawing S8225(7)-1428724 shall be provided. The inflatable floor shall be constructed of coated cloth conforming to type 6 of Specification MIL-C-17415 reinforced on both sides with coated cloth conforming to type 1 of Specification MIL-C-17415. The inflatable floor shall be designed for inflation to a working pressure of 2 p.s.i. Six tie patches shall be provided for securing the floor to the boat bottom. The floor shall be securely fastened in place before the boat is packed. Two inflatable life preservers, furnished as specified in 3.3.1, shall be securely fastened with 1/8 inch diameter nylon line to the inflatable floor tie loops at the after end, one to the starboard side the other to the port side. The life preservers shall be removed from the pouch and laid out flat. A boarding handle shall be fitted as shown on Drawing S8225(7)-1428724.

3.6.7 Arch tube. - An arch tube shall be provided as shown on Drawing S8225(7)-1428724. The arch tube shall be inflated simultaneously with the main tube, and from the after end. This shall be accomplished through an opening in the end of the arch tube. Molded collars may be used for the ends of the arch tube if the contractor so desires.

3.6.8 Light. - A light in accordance with Specification MIL-L-22686 shall be provided and installed. A 1/8 inch diameter braided nylon line approximately 12 inches long shall be attached to the tear tab and to the release plug in the stowage case.

3.6.9 Inflation system. - The inflation system shall consist of a CO_2 cylinder, charged with sufficient CO_2 to inflate the boat to a minimum of 2-1/2 p.s.i. and a maximum of 3 p.s.i. pressure at 70° Fahrenheit (F.) in 1 hour. The cylinder shall be fitted with a drop type valve which shall be connected to a manifold by a flexible hose connection. The design of the inflation system shall be such that the two main compartments of the lifeboat shall attain the designed shape in approximately 30 seconds at approximately 70° F., the time being measured from the opening of the CO_2 valve. The CO_2 cylinder shall be installed so that the actuating cable of the cylinder valve shall lead to the pull cable plug on the stowage case. The actuating cable for the CO_2 cylinder valve shall be in accordance with Drawing 645-1428951. The cable shall be secured with a 10-pound (breaking strength) wire seal to the outlet ferrule on the valve to prevent accidental inflation. Six additional wire seals shall be provided (see 3.6.19). The assembled actuating cable shall withstand a pull of 100 pounds.

3.6.9.1 Cylinders. - The cylinders shall be fitted with siphon tube designed so that the discharge of carbon dioxide will be complete and rapid with the cylinder in any intermediate position between the vertical, with the valve up, and the horizontal. The following information shall be stencilled in letters 1/2 inch high, with durable black paint, on the side of each cylinder:

Tare weight . . . . . pounds
Weight of CO_2 . . . . . pounds
For lifeboat, man overboard, CO_2 inflatable

3.6.9.2 Cylinder valves. - The cylinder valves shall comply with the requirements specified herein and shall conform to the applicable requirements of Specification MIL-I-18369. The design of the valve shall conform with the detail drawings and specifications provided by the manufacturer, except when conflicting requirements exist, in which case this specification shall apply. All parts other than seat materials and packings shall be made of brass, bronze, or corrosion-resisting steel, and shall be designed to operate freely under all conditions. The valve shall provide an unrestricted passage for the discharge of gas to the distributing device without leakage, and shall be proportioned to prevent freezing or accumulation of snow when operated at temperature of minus 20°F. The valve shall be as compact as possible and the distance from the top of the cylinder to the top of the valve (including rigid attachments) shall not exceed 3-1/2 inches. The maximum weight for a complete valve assembly shall not exceed 1.7 pounds. The force required to operate the valve and discharge the contents shall not exceed 10 pounds at normal temperature, when measured on a tensile machine operating at a rate of 12 inches per minute. The cables shall not jam in the valve when pulled through the actuating cycle in a tensile machine at the rate of 12 inches per minute with a constant resistance of 50 pounds applied counter to valve sheave rotation. The maximum pull with the wire seal in position shall be 13 pounds. The length of pull for complete operation of the valve and release of the cable shall be approximately

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3-1/4 inches. The valve shall be of such design that the direction of pull can be either down toward the cylinder or out away from the cylinder. The modification of the valve shall be such that it will be possible to change from one direction of pull to the other with the use of an open end wrench and without removal of the valve from the cylinder. The direction of pull for this boat shall be down toward the cylinder. The design shall be such that the cable assembly can be attached to the valve after all other connections are made up in order to eliminate the possibility of discharging the cylinder accidentally while the unit is being handled. The outlet of the valve shall be of the nonrecoiling type having a male 3/8 inch - 18 American National Straight Pipe Thread in accordance with Handbook H28 for attachment to the flexible hose of the manifold. Recharging of the cylinders after a normal operation shall be possible without the use of recharge bonnets and without the need for any replacement parts. The valve shall incorporate a safety device operating on the principle of the frangible disk and designed to rupture at a pressure of 3150 to 3500 p. s. i. This disk shall be in accordance with piece No. 22 of Drawing 645-1428951. Discharge in the event of rupture of safety disk shall be directly into the ambient air and not into the lifeboat, and should be regulated by discharge holes having an aggregate area of 0. 00475 square inch, so placed as to eliminate recoil of the cylinder.

3. 6. 9. 3 Flexible hose. - The connection from the cylinder valves to the manifolds shall be accomplished by means of a flexible hose and the necessary fittings. The hose and end fittings shall be in accordance with Drawing 645-1428951. The pull cable shall also be enclosed in a flexible hose. The hose shall be secured by fabric loops. The pull cable hose shall be Walter Kidde Company Part No. 842127 or equal.

3. 6. 9. 4 Manifold. - Each boat shall be fitted with a brass manifold which shall perform the functions and include the fittings specified in (a) through (c):

(a) Be so arranged to assure simultaneous and approximately equal inflation of the two compartments to which they are attached.
(b) Have a nipple suitable for attachment of the fittings on the delivery end of the flexible hose.
(c) Be fitted with nonreturn valves to prevent the passage of the inflation medium from one compartment to the other or to the open air.

3. 6. 9. 4. 1 The manifold shall project a minimum distance from the outer surfaces of the boat in order to form as small projections as possible. The manifold shall be set in a molded neoprene base properly secured to and forming an integral part of the tube. All portions of the manifold which may project into the main tube shall be covered and there shall be no rough or sharp edges to cut or injure the inside surfaces of the main tubes or bulkhead when the boat is deflated and folded. All exposed metal parts of the manifold shall be covered with molded neoprene in a manner approved by the bureau or agency concerned to prevent damage to the boat fabric. Drawings of manifolds shall be submitted to the bureau or agency concerned for approval.

3. 6. 9. 5 Inflation valves. - All inflation valves shall be located as shown on Drawing S8225(7)-1428724. The inflatable floors shall be fitted with neoprene base valves, Peters and Russell, Inc. No. 3585 or equal. The main tubes shall be fitted with neoprene base valves, Peters and Russell, Inc. No. 3485, or equal. One extra valve insert per boat shall be furnished. The valve insert shall have six screws fastened to the insert with a gasket, 0. 3 inch minimum thickness, conforming to class 2 of Specification MIL-A-7021 (see 3. 6. 19). The inflation valves are for the purpose of inflating, deflating or topping-off by a hand pump.

3. 6. 10 Pockets. - All pockets shall be clearly and indelibly labeled to indicate the contents thereof. All pockets shall be attached as shown on Drawing S8225(7)-1428724.

3. 6. 10. 1 Pump pocket. - A pocket shall be provided for the hand pump furnished (see 3. 3. 9. 3).

3. 6. 10. 2 Sealing clamp pocket. - A pocket for the repair items shall be provided. These pockets shall be provided with fasteners (see 3. 3. 5). The pocket shall be attached to the main tube along its entire length, and for at least two inches in width. The attachment shall be reinforced along the top and bottom edges by coated cloth 1-5/8 inches wide conforming to type 8, class B of Specification MIL-C-17415, cemented to the tube and to the pocket to form a V-shaped crotch strip (see 3. 6. 19).

3. 6. 11 "Dee" rings. - One and one fourth "Dee" rings conforming to Drawing 645-1428951 shall be fitted as shown on Drawing S8225(7)-1428724.

3. 6. 12 Lines. -

3. 6. 12. 1 Life lines. - Life lines shall be fitted with slack around the periphery of the boat as shown on Drawing S8225(7)-1428724.

3. 6. 12. 2 Righting and boarding handles. - Righting and boarding handles shall be installed as shown on Drawing S8225(7)-1428724.

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3.6.13 CO_2 cylinder holder. - A suitable CO_2 cylinder holder shall be provided and installed as shown on Drawing S8225(7)-1428724. The cylinder holder shall be fitted with nylon drawstrings or straps around the base of the neck and straps around the bottom of the cylinder. The cylinder holder shall be sufficiently rugged to withstand the forces developed while the CO_2 cylinder is being discharged and when the lifeboat is dropped from a height of 50 feet. The cylinder holder shall be so located that when the CO_2 cylinder is in place, the hose between the cylinder and manifold shall not be distorted when the boat is in a packed or inflated condition.

3.6.14 "Dee" ring patches. - "Dee" ring patches shall be fitted as shown on Drawings S8225(7)-1428724.

3.6.15 Life line patches. - Life line patches shall be installed around the periphery of the boat as shown on Drawing S8225(7)-1428724. Each patch shall withstand a 250-pound pull exerted in a direction perpendicular to the base of the patch.

3.6.16 Stowage case. - Each lifeboat shall be packed in a stowage case. The stowage case shall be in accordance with Drawing 645-1973948.

3.6.17 Sea anchor. - A sea anchor and line in accordance with Drawing S8225-1298540 shall be provided. The sea anchor line shall be secured to the "Dee" ring with a fishermans bend and the free end seized. The sea anchor and line shall be so stowed in the folded boat that it will stream without fouling when the boat is inflated.

3.6.18 Operation and maintenance manual. - Each lifeboat shall be furnished two copies of an operation and maintenance manual 6 by 9 inches, printed on waterproof paper. One copy of the manual shall be placed in the sealing clamp pocket. The other copy shall be sealed in a coated cloth envelope which shall be cemented to the outside of the stowage case near the hand grip. This envelope shall have stencilled in large letters on the outside "Operation and Maintenance Manual". Twenty additional copies of the manual shall be furnished the bureau or agency concerned.

3.6.19 Equipment. - The contractor shall furnish the following items of equipment with each lifeboat, except the two inflatable life preservers which will be furnished as specified in 3.3.1.

Item		Quantity
(a)	Hand pump (see 3.3.9.3)	1
(b)	Valve insert with 6 screws (see 3.6.9.5)	1
(c)	Spare CO_2 cylinder	1
(d)	Life preservers (see 3.3.1)	2
(e)	Wire seals (see 3.6.9) packed in water-resistant envelope	6
(f)	Adapter for valve (see 3.3.9.3)	1
(g)	Sealing clamp, 3 inches, Drawing 645-1428951	1
(h)	Sealing clamp, 5 inches, Drawing 645-1428951	1
(i)	Sealing clamp, 7-1/2 inches, Drawing 645-1428951	1
(j)	Operation and maintenance manual (see 3.6.18)	2
(k)	Sea anchor and line, Drawing S8225-1298540	1

NOTES:
1. Items (g), (h) and (i) shall be securely bound together and inclosed in a 4 mil thick, heat sealed, polyethylene container, with air removed to provide compact package, before being placed in the sealing clamp pocket.
2. Items (b), (e) and (f) shall be wrapped in transparent material and inclosed in a 4 mil thick, heat sealed polyethylene container and stowed in the sealing clamp pocket.

3.7 Marking. -

3.7.1 Instructions. - Instructions relative to topping-off and deflating shall be provided on the flotation tube near the topping-off valves. Instructions relative to topping-off and deflating shall read as follows:

"TO INFLATE - Insert pump with adapter into valve. Open valve to left. Pump until firm. Close valve to right.
TO DEFLATE -Turn valve to left."

3.7.1.1 The letters shall be of suitable size so as to be clearly legible. Instructions specified for location on the boat, shall be printed in black letters on a white rubber patch, or shall be printed, in white, directly on the tube coated cloth by the transfer system provided the method used is permanent for the life of the boat, or shall be in the form of a decal which has been approved by the bureau or agency concerned. Boarding station and arrow shall be marked on both sides of the main tube.

3.7.2 Labeling. - Each boat shall have fitted on the top of the main tube, at the stern, a molded patch, or an equivalent label approved by the bureau

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or agency concerned, containing the following information:

"Lifeboat, man overboard, CO inflatable,
Stock No. _____
Built for the United States Navy
Contractor, city, month, year
Contract or order number
Registry No."

3.7.2.1 The registry numbers assigned to these boats will be furnished by the contracting officer upon request.

3.7.3 Folding instructions. - Each lifeboat, complete with all accessory equipment stowed in its proper place shall be lightly dusted with soapstone or talcum and folded in accordance with (a) through (h) for stowage in the stowage case.

(a) Deflate all sections of the boat floor and main tube. Close all valves tightly. Place fully charged CO_2 cylinder in holder and connect to manifold. Pass pull cable housing through holder loops. All sections of the boat shall be laid out as flat as possible (see step 1 of figure 1). Attach sea anchor line to bow "Dee" ring and stow sea anchor so that it falls into the water upon inflation of the boat. The anchor line shall be folded into a 12 inch bank and tied in two places by three wraps of 30/6 cotton thread. Unfold both preservers. Secure one end of a 10 foot length of 1/8 inch diameter commercial braided nylon cord to the waist strap of each preserver. Tie one preserver line to the port side forward tie-loop and the other to the starboard side aft floor tie-loop. Lay preservers flat on floor and under tube fold adjacent to respective tie loops. All exposed metal surfaces which may chafe the boat fabric in the folder condition shall be suitably and securely covered with tape, tubing or waste fabric strips.
(b) Fold bow end toward stern of boat reducing length by approximately one third. Stow sea anchor on top and in center of folded section (see step 2 of figure 1).
(c) Fold stern end toward bow of boat over the above fold (see step 3 of figure 1). Lay cylinder and sea light actuating cables towards stern end.
(d) Fold port side reducing width by approximately one third (see step 4 of figure 1).
(e) Fold starboard side over the above fold (see step 5 of figure 1).
(f) Place folded boat (see step 6 of figure 1) into lower half of rigid case (cylinder on bottom of case) with cylinder and sea light actuating cables facing case plug. Connect both cables to case plug assembly.
(g) Place upper half of case over lower half. Be sure case halves are properly aligned to make water tight seal.
(h) Secure rubber studs into mating holes of break-away assembly and tighten adjusting straps.

3.8 Performance. - Lifeboats and component items shall meet the performance tests specified in 4.7.

3.9 Workmanship. - The workmanship shall be in accordance with high-grade manufacturing practice covering this class of work. Because of the emergency use of this equipment, it is required that all details of workmanship be of the highest quality in order to insure proper functioning of the lifeboat at all times. Defects in workmanship shall constitute cause for rejection.

3.10 When any specification which forms a part of this specification requires that the product be subjected to and pass qualification tests, only products which are listed on the applicable Qualified Products List on the date of invitation for bids or which may be added to that Qualified Products List subsequent to that date shall be utilized in the construction of equipment specified to be in accordance with this specification. In the event no Qualified Products List has been issued, the contractor shall request instructions as to what testing will be required to determine whether the product meets the requirements of the specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection. - The examination and testing of lifeboats shall be classified as follows:

(a) Qualification inspection. - Qualification inspection is the inspection performed on samples submitted for approval as qualified products.
(b) Preproduction tests. - Preproduction tests consist of all the tests specified herein and shall be performed on models representative to the production item after the award of a contract to determine that the production items meet the requirements of this specification.
(c) Acceptance inspection. - Acceptance inspection is the inspection performed in accordance with 4.6.

4.2 Qualification inspection. 1/ - Qualification inspection shall be conducted at a laboratory

1/ Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.2.1).

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satisfactory to the Bureau of Ships. The inspection performed on the completed lifeboat and component material shall consist of all tests specified in this specification, including examination to determine strict conformance to the construction requirements. Samples of all material used in the fabrication of all lifeboats shall be examined and tested to ascertain their conformance to the applicable specifications or the requirements for the material as specified herein.

4.2.1 Samples for qualification inspection. - The following samples shall be submitted for qualification inspection:

One lifeboat complete with equipment and carrying case.
Two yards of all coated cloth used (vulcanized).
Two yards of all cloth used (uncoated).
Five samples of a 5-inch diameter life line patch cemented and vulcanized to a 7-inch diameter piece of type 16, class A coated cloth in accordance with Specification MIL-C-17415.
Two sample "Dee" ring patches vulcanized to an 18 inches by 14 inches piece of type 16, class A cloth in accordance with Specification MIL-C-17415.
Two vulcanized seams 7 inches wide by 12 inches long of type 16, class A cloth in accordance with Specification MIL-C-17415. Seams are to be prepared and vulcanized in the same manner as the boat tube seams.
Five yards of each size of nylon rope used.

4.3 Preproduction sample. - Prior to starting regular production the contractor shall submit the following samples:

One lifeboat complete with equipment and carrying case.
Two yards of all coated cloth used (vulcanized).
Two yards of all cloth used (uncoated).
Five samples of a 5-inch diameter life line patch cemented and vulcanized to a 7-inch diameter piece of type 16, class A coated cloth in accordance with Specification MIL-C-17415.
Two sample "Dee" ring patches vulcanized to an 18 inches by 14 inches piece of type 16, class A cloth in accordance with Specification MIL-C-17415.
Two vulcanized seams 7 inches wide by 12 inches long of type 16, class A cloth in accordance with Specification MIL-C-17415. Seams are to be prepared and vulcanized in the same manner as the boat tube seams.
Five yards of each size of nylon rope used.

4.3.1 Preproduction test samples shall be forwarded to the Commander, New York Naval Shipyard Material Laboratory (947) Brooklyn, N. Y., plainly identified by securely attached durable tags marked with the following information:

Sample for preproduction test.
Name of product, together with complete identifying data such as manufacturer's designation number.
Name of manufacturer.
Contract or order number.

4.4 Contractor's responsibility. -

4.4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.4.2 The following tests shall be conducted on each lifeboat by the contractor under the surveillance of the Government inspector in the order listed. Lifeboats not meeting the requirements of these tests shall be corrected prior to submitting the lifeboats for acceptance. Treating or swashing the inflatable member of the lifeboat with latex or any other liquid sealing agents in order to pass these tests will not be permitted. It is the contractor's responsibility to recharge the CO_2 cylinders discharged during test. Only cylinders, valves and pull cables which have actually been used during the tests are to be delivered with the boats. This is not intended to restrict the cylinder assembly to the boat on which it was tested.

Test	Paragraph
Inflation	4.7.1
Overpressure leakage	4.7.2
Working pressure leakage	4.7.3

4.4.2.1 Wherever, prior to acceptance, repair or reworking is found to be necessary, it shall be performed by the hot spot cure or other method acceptable to the Government inspector with consideration being given to type of defect being repaired. A maximum of five repairs per boat including bulkhead repairs may be made. No area shall be cured more than twice. Only one bulkhead repair is permissible on any boat and not more than 5

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percent of the boats under the contract shall have a bulkhead repair. Both an inner and an outer fabric patch will be used to repair tears or cuts. The decision as to what constitutes repairable defects shall be the responsibility of the Government inspector.

4.4.3 The contractor shall prepare a minimum of 5 adhesion test samples from each day's production and shall subject the samples to the tests specified in 4.7.5. The dimensions of the test specimen shall be as specified in 4.7.5. The seam samples shall be cured simultaneously with a boat or group of boats.

4.5 Test conditions. - Unless otherwise specified herein, all tests required by this specification shall be conducted under the standard atmospheric conditions specified in Specification CCC-T-191. Samples shall be conditioned in the standard atmosphere for at least 24 hours before testing.

4.5.1 Temperature correction. - If the ambient temperature changes during the air-pressure tests adjustment shall be made on the basis of a change in pressure of 0.031 p.s.i. per degree F. change in temperature.

4.5.2 Barometric pressure correction. - If the barometric pressure changes during air pressure tests, adjustment shall be made on the basis of change in pressure of 0.49 p.s.i. per inch (mercury) change in barometric pressure. An adjustment of 0.10 p.s.i. for each 1/5 inch (mercury) change in barometric pressure may be made at the discretion of the contractor.

4.5.3 Pressure measurement. - Pressure shall be measured by means of a mercury manometer.

4.6 Sampling for lot acceptance inspection. -

4.6.1 Inspection lot. - All lifeboats submitted at one time shall be considered a lot for the purpose of acceptance inspection.

4.6.2 Sampling for examination. - A random sample of lifeboats shall be selected from each lot in accordance with inspection level II of Standard MIL-STD-105 for the examination specified in 4.6.4. The acceptable quality level shall be 1.0 for major defects and 2.5 for minor defects per 100 units. If any critical defect is found in the sample boats the manufacturer's inspection procedures shall be evaluated again and every boat shall be reinspected for all critical defects.

4.6.3 Sampling for lot acceptance tests. - A random sample of lifeboats shall be selected in accordance with table I and subjected to the tests specified in 4.7.1, 4.7.2 and 4.7.3. If any one of the lifeboats in the sample is rejected, the entire lot represented shall be rejected.

Table I - Sampling for lot acceptance tests.

Number of boats in inspection lot	Number of boats in sample	Acceptance number (defectives)	Rejection number (defectives)
8 and under	All	-	-
9 to 15	8	0	1
16 to 25	10	0	1
26 to 40	15	0	1
41 to 65	20	0	1
66 to 110	25	0	1
111 and over	35	0	1

4.6.4 Examination. - Each completed sample lifeboat selected in accordance with 4.6.2 shall be carefully examined to determine conformance with this specification, inspected for the fitting of the protection liners in way of manifold and subjected to such additional inspection as may be considered necessary to insure suitability for the purpose intended. Defects are described and classified in table II.

4.7 Test methods. -

4.7.1 Inflation test. - The lifeboat complete in its carrying case shall be inflated in accordance with the instructions on the instruction flap located on the carrying case to ascertain that the case will open properly and the inflation system performs properly. The lifeboat shall inflate to its design shape in not more than 30 seconds at 70° F., the time being measured from the instant the CO_2 inflation valve is discharged. The boat shall be allowed to stand for 1 hour to allow the gas in the tubes to come to room temperature. Pressure readings shall be taken through the valves of each compartment. Pressure in both compartments shall be between 2 and 3 p.s.i. when the temperature is adjusted to approximately 70° F. as specified in 4.5. Failure of the stowage case to open properly or failure of inflation system to operate properly shall be cause for rejection.

4.7.2 Over pressure leakage test. - Upon completion of the inflation test as specified in 4.7.1, the pressure in the lifeboat shall be increased to 5 p.s.i. and allowed to stand for 10 minutes. If the pressure at the end of the 10 minute period is less than 4.75 p.s.i., the lifeboat shall be rejected. The inflatable floor shall also be subjected to the above test and if the pressure at the end of the 10 minute period is less than 4.75 p.s.i. the lifeboat shall be rejected. Upon completion

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of this test, the lifeboat shall immediately be subjected to the test specified in 4.7.3.

4.7.3 Working pressure leakage test. - One compartment of the flotation tube and the inflatable floor shall be inflated by air to 2.0 p.s.i. through the plug valves, and allowed to stand for 1 hour. The pressure shall then be checked and readjusted as necessary to the original pressure of 2.0 p.s.i. After standing 24 hours, the pressure shall be measured. The main tube compartment under test shall be deflated and the same test conducted on the remaining compartment. If any compartment has a corrected pressure of less than 1.25 p.s.i. or if any cross tube or floor has a corrected pressure of less than 1.20 p.s.i. after 24 hours, the lifeboat shall be rejected.

Table II - Classification of defects.

Categories	Defects
Critical:	
1	Coated cloth in inflatable compartments damaged or not properly cured.
2	CO_2 cylinder to manifold hose damaged, or end connections not tight.
3	Tube seams not taped or gum stripped as specified.
4	CO_2 protection liners missing or not properly installed.
5	CO_2 cylinder holders missing or not properly installed and located.
6	Release cable damaged or not properly installed.
Major:	
101	Coated cloth in noninflatable compartments and patches damaged or not properly cured.
102	Hand pumps and adapters missing, inoperable, or not properly installed.
103	Repair kit missing, incomplete, or contains damaged or incorrect components.
104	"Dee" rings will not swivel, not brass, or not properly installed.
105	Snap and lift fasteners damaged, or not properly installed and located.
106	Metal parts not coated as specified.
107	Rope and lines damaged, or ends not fused as specified.
108	Thread used in construction not as specified.
109	Radar reflective screen missing, nonconforming or not stowed as required.
110	Evidence of sewing used in seam or through coated cloth or air chamber.
111	Attachment seam between bottom and lower tube not reinforced as specified.
112	Life lines missing, damaged, or not properly installed.
113	Ends of arch tubes not reinforced as specified.
114	Righting and boarding handles missing, damaged not properly installed or located.
115	Sea anchor "Dee" ring missing, or not properly installed and located.
116	Inflation valves missing, damaged, or not properly installed or closed.
117	Instruction manuals missing.
118	Instruction lettering missing, incorrect, or illegible.
119	Length and width of boat (inflated) not as specified.
120	Tube diameters not as specified.
121	Airtight bulkheads missing, damaged, or not properly located.
122	Salt water light not properly installed.
123	Evidence of unauthorized or nonconforming patching repair or rework; number of repairs per boat exceeds allowable maximum; area cured more than allowable number of times.
124	Evidence of treating or swashing inflatable members with latex or other liquid sealing agents
125	Sea anchor missing or not as specified
126	Boat improperly folded
Minor:	
201	Pockets not reinforced as specified.
202	Pockets not securely sealed as specified.
203	Identification marking missing, incorrect, or illegible.
204	Nylon rope ends not smoothly fused
205	Inside width of boat (inflated) not as specified.
206	Markings on hand pumps missing, incorrect, or illegible.
207	Pump and sealing clamp pocket fasteners inoperable.
208	Packing not as specified.

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4.7.4 Drop test. - One lifeboat, complete with all Government and contractor furnished equipment shall be set for operation and dropped into water from a height of 50 feet. Immediately after the drop, the boat shall be inflated by the CO_2 inflation system to ascertain that the boat functions properly. The lifeboat shall then be tested as specified in 4.7.3.

4.7.5 Seam tests. - Five 2-inch wide seam specimens shall be tested for tensile strength at a machine speed of 12 inches per minute.

4.7.6 Life line patch tests. - Life line patches shall be tested for tensile strength in a suitable jig. The direction of the pull shall be perpendicular to the patch.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Packaging. -

5.1.1.1 Each stowage case (see 3.6.16) complete with folded lifeboat and equipment (see 3.6.19) except spare CO_2 cylinder will require no further packaging.

5.1.1.2 Each spare cylinder (see 3.6.19(c)) shall be packaged in a fiberboard box conforming to class 2, of Specification PPP-B-636. Type and style of box shall be at the option of the contractor. All center and edge seams and manufacturer's joint of the box shall be water proofed with tape in accordance with the appendix to the box specification.

5.1.2 Packing. - Each stowage case complete with folded lifeboat, equipment and a spare cylinder shall be packed in an exterior shipping container in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the first receiving activity for early installation. The shipping container shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.3 Marking. - Shipment marking information shall be provided on the exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, registry number, contract or order number, contractor's name and destination. The registry number does not apply to the spare CO_2 cylinder.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Packaging. -

5.2.1.1.1 Levels A and C. - Packaging shall be as specified in 5.1.1.1 and 5.1.1.2.

5.2.1.2 Packing. -

5.2.1.2.1 Level A. - Each stowage case, with lifeboat and equipment, and a spare cylinder shall be packed in containers conforming to any of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-576	Class 2
PPP-B-585	Class 3 use
PPP-B-636	Class 2
PPP-B-640	Class 2

Shipping containers shall be closed, strapped or banded in accordance with the appendix to the applicable box specification.

5.2.1.2.2 Level B. - Each stowage case, with lifeboat and equipment, and a spare cylinder shall be packed in containers conforming to any of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-576	Class 1
PPP-B-585	Class 1 or 2 use
PPP-B-636	Class 1
PPP-B-640	Class 1

Shipping container closure shall be in accordance with the appendix to the applicable box specification.

5.2.1.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. Marking information on shipping containers shall include the registry number.)

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging, packing and marking of other than as specified in 5.1 (see 5.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 23007, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2.1).

6.2.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or any other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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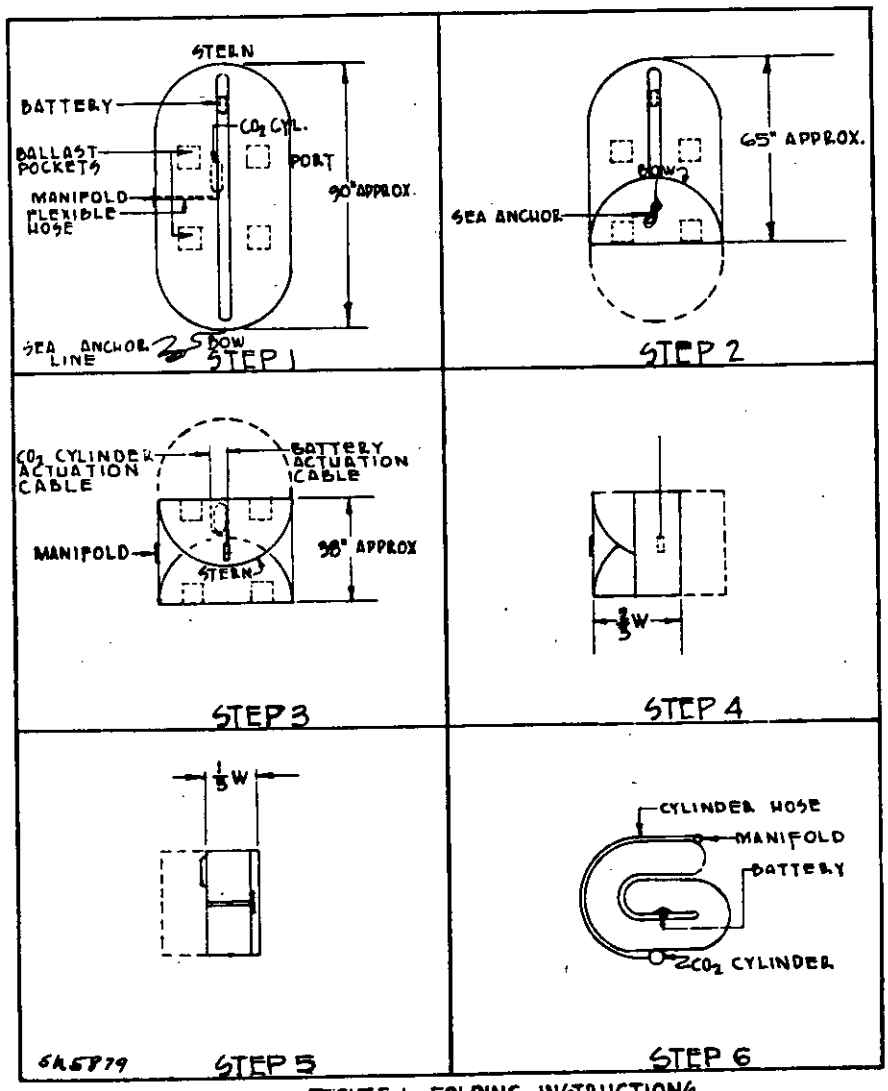


FIGURE 1 - FOLDING INSTRUCTIONS

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